AI FOR CREATIVITY · LEVEL 6 · FUTURE EDUCATION UNIVERSITY COLLEGE

R&D Skills Workbook

Research, research gaps, datasets and UCA Harvard referencing — exercises for the Research Skills Workshop

Exercise	Skill	Slides
1	Build a search strategy	Finding sources
2	Evaluate a source (CRAAP test)	Judging quality
3	Build a literature matrix	Synthesis
4	Spot the gap — name the type	Research gaps
5	Write a problem statement and research questions	From gap to question
6	Evaluate a dataset	Datasets
7	Draft your datasheet	Describing data
8	In-text citations	UCA Harvard
9	Fix the reference list	UCA Harvard
Key	Answer key released on Moodle after the workshop	

Name

Student name: _____ Project idea (one line): _____

Exercise 1 · Build a search strategy

Write your topic as a question, break it into 2–3 concepts, then list synonyms for each. Combine with AND (between concepts) and OR (between synonyms). Use quotation marks for exact phrases.

	Concept 1	Concept 2	Concept 3
Worked example	"Al Sadu" OR "Sadu weaving" OR "Bedouin textile"	"deep learning" OR "computer vision" OR "image classification"	authentic* OR counterfeit OR imitation
My concepts			
My synonyms			

My final search string:

Database	Date searched	Results	Screened	Kept
Google Scholar				
UCA library search				
arXiv / CVF Open Access / ACM DL				
Other (museum, industry, dataset sites)				

Tip: the asterisk (authentic) finds authentic, authenticity and authentication. Record every search — this log becomes evidence of your research process.*

Exercise 2 · Evaluate a source with the CRAAP test

Pick one source you found. Score each criterion 1 (weak) to 3 (strong) and give a reason.

Criterion	Ask yourself	Score 1–3	Reason
Currency	When was it published? Is that recent enough for an AI topic?		
Relevance	Does it address my question, or only the general topic?		

Criterion	Ask yourself	Score 1–3	Reason
Authority	Who wrote it? Peer-reviewed journal, conference, university, respected institution?		
Accuracy	Is it supported by evidence, data and references? Can I verify it?		
Purpose	Is it trying to inform, sell, or persuade? Any bias?		

The CRAAP test was developed by the Meriam Library, California State University, Chico.

Exercise 3 · Build a literature matrix

Fill in one row per source. The last column is where your research gap will appear.

Source (Author, Year)	Theme	Method / data used	Key finding	What it does NOT cover for my project

Now write two sentences that **synthesise** (connect) at least two sources, e.g. 'While A (2014) shows…, B (2021) warns that…; neither considers…'

Exercise 4 · Spot the gap

For each statement, name the type of gap: **Context · Data · Method · Application · Evidence · Ethical**.

#	Statement from a student's review	Gap type
1	Most music-generation studies train on Western classical and pop; very few include Khaleeji or Arabic maqam music.	
2	I found no public dataset of labelled Arabic calligraphy styles with licences that allow model training.	
3	Previous studies of AI fashion recommenders only measured click-through rates, never whether designers found the suggestions creatively useful.	

#	Statement from a student's review	Gap type
4	Style-transfer tools are widely studied in fine art, but I found no studies of their use in architectural presentation boards.	
5	Claims that AI image tools save designers time are mostly from company blogs; I found no independent studies measuring it.	
6	Papers on AI voice cloning rarely discuss how the original performers consented or were paid.	

Exercise 5 · Problem statement and research questions

Formula

Problem statement = WHO is affected + WHAT is wrong + WHY existing work does not solve it (your gap).

Good research question = specific · answerable with your method and data · achievable in the time · not just yes/no.

My problem statement:

My aim (one sentence, starting 'To…'):

Improve these weak research questions:

Weak question	Why it is weak	My improved version
Can AI help fashion?		
Can AI recognise Arabic calligraphy?		
Will my app stop fake artworks being sold in the UAE?		

Exercise 6 · Evaluate a dataset

Choose a dataset you might use (e.g. from Kaggle, Hugging Face, a museum open-access collection or your own collection). Answer the questions, based on Gebru et al.'s (2021) 'Datasheets for datasets'.

Question	My answer
Who created it, when and why?	
What is in it? (classes, number of items, format, resolution)	
How was it collected? Did people consent?	
What licence applies? Can I use it for training? Commercially?	
How was it labelled, and by whom?	
Who or what is missing or under-represented?	
Could any items appear in both training and test sets (leakage)?	

Red flags

No licence stated · images scraped from social media · faces or personal data without consent · one class much larger than others · near-duplicate images · labels you cannot check.

Exercise 7 · Draft your datasheet

This table becomes Section 5 of your R&D document. See the exemplar for a completed version.

Field	My project
Name and version	
Purpose	
Classes and counts	
Sources (with permissions)	
Collection method	
Labelling (who, how checked)	
Pre-processing	

Field	My project
Splits (and how you avoid leakage)	
Licence and sharing	
Known limitations and bias	
Storage and privacy	

Exercise 8 · In-text citations (UCA Harvard)

UCA rules

Author and year in brackets: (Boden, 2004). With a direct quotation add the page: (Butler, 2006:8).

Two authors: (Howard and Gugger, 2020). Three or more: (Gebru *et al.*, 2021).

Author named in your sentence: Boden (2004) argues…

Secondary citation: (Original Author, Year:page cited in Later Author, Year:page).

Source: UCA Library (n.d.) *UCA Harvard referencing.* At: https://mylibrary.uca.ac.uk/referencing (Accessed 23/09/2026).

Write the correct in-text citation for each situation:

Situation	Your citation
a) Paraphrasing Kate Crawford's 2021 book	
b) Quoting page 45 of Margaret Boden's 2004 book	
c) Paraphrasing the ResNet paper by He, Zhang, Ren and Sun (2016)	
d) Starting a sentence with the fastai book by Howard and Gugger (2020)	
e) Quoting Lovelace (1843), which you read quoted on page 12 of du Sautoy (2019) (hypothetical page)	

Exercise 9 · Fix the reference list

Each reference has at least one error against the UCA Harvard style. Rewrite it correctly.

#	Incorrect reference	Corrected reference
1	The creative mind, Margaret Boden, Routledge, 2004	
2	Gebru, T. et al. (2021) Datasheets for datasets. Communications of the ACM.	
3	https://www.robots.ox.ac.uk/~vgg/data/dtd/	
4	Howard, J. & Gugger, S., Deep Learning for Coders (O'Reilly)	

#	Incorrect reference	Corrected reference
5	ChatGPT (2026) "how do I fix my error"	
6	Buolamwini and Gebru. Gender Shades. 2018.	